Business Requirements Document (BRD) & Technical Assessment: AI Solutions Engineer

Company Context: ABC Company (Automotive OEM Division)
Role Target: AI Solutions Engineer / AI Integration Specialist
Assessment Duration: 6 hours (Take-Home Technical Project)

1. Executive Summary & Objective

The objective of this assignment is to evaluate a candidate's ability to design, build, and integrate an end-to-end conversational AI Agent (Chat or Voice) tailored for an enterprise Automotive Original Equipment Manufacturer (OEM) like Mahindra & Mahindra. The agent must dynamically recognize the customer's intent across four distinct stages of the automotive purchase and ownership lifecycle, maintaining real-time bidirectional synchronization with Zoho CRM.

2. Comprehensive Customer Lifecycle Matrix

Lifecycle Stage	User Context & Trigger	Expected AI Agent Logic & Response	Zoho CRM Module & Workflow
1. New Lead	Unidentified visitor asking about vehicle models (e.g., XUV700, Thar, Scorpio-N), variant pricing, or features.	Answers specs/pricing query, pitches test drive booking, collects Full Name, Phone, Email, and Preferred City.	Create a new entry in **Leads** module with vehicle model in Custom Field.
2. Ongoing Pipeline	Existing prospect providing phone number or deal ID to check test drive confirmation, quotation, or dealer contact.	Fetches active deal status from CRM, informs customer of scheduled time/quote, and updates follow-up preferences.	Search & update record in **Deals / Potentials** module.
3. Booked Vehicle	Customer who paid booking amount asking for delivery timeline, VIN allocation, or payment	Validates Booking ID or Phone, provides current allocation stage (e.g., In Transit, Dispatch	Retrieve record from custom module / Deal stage marked as **"Closed Won - Booking**

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	link for balance.	Pending, Delivered).	**Done"**.
4. Post-Purchase / Service	Existing owner logging a complaint, asking about service intervals, or booking a periodic maintenance slot.	Gathers registration number, odometer reading, reported issue/service type, and preferred service center location.	Create record in **Cases / Service Tickets** module linked to existing Contact.

3. Technical Options & Requirements

Candidates have to complete **Multistage AI Chat Agent** for their project submission.

Multistage AI Chat Agent

- **Interface:** Build a responsive frontend interface using React, or Next.js.
- **LLM Integration and Orchestration:** Implement tool/function calling to integrate the LLM with backend services, and support multi-step agent workflows where required, using LangChain, LlamaIndex, or direct integration with LLM provider APIs/SDKs.
- **Free/Open Source AI Models:** Utilize free API tiers (e.g., Groq API with Llama-3/DeepSeek, Google Gemini API free tier, or local HuggingFace models via Ollama).
- **Zoho CRM Integration:** Connect using Zoho CRM OAuth 2.0 REST API or Webhooks to search, create, and update records across Leads, Deals, and Cases.
- **State Management:** Maintain conversational history and session state to seamlessly switch context between vehicle discovery and ticket creation.

4. Setup Instructions for Free Developer Tier (Zoho CRM)

1. Sign up for a free developer account at https://www.zoho.com/crm/.
2. Go to **Zoho API Console** (developer.zoho.com) and create a **Self Client** application.
3. Generate authorization code with scope: ZohoCRM.modules.ALL,ZohoCRM.settings.ALL.
4. Generate refresh_token, client_id, and client_secret to manage Access Tokens inside

your backend service.

5. Pre-populate the CRM with mock data:
    - 1 Existing Lead (e.g., Rajesh Sharma - Interested in Thar)
    - 1 Active Deal (e.g., Priya Patel - Test Drive Scheduled for XUV700)
    - 1 Booking Record (e.g., Booking ID #MAH-9921 - Scorpio-N Z8L)

5. Candidate Submission Checklist & Deliverables

- **Private GitHub Repository:** Clean, modular codebase containing README.md with execution steps, setup scripts, and .env.example.
- **Architecture Diagram:** System workflow diagram illustrating intent classification, tool calling, and CRM API integration.
- **Demo Video (5–7 Minutes Loom):** Walkthrough demonstrating all 4 stage scenarios in action along with real-time updates inside Zoho CRM dashboard.

6. Candidate Evaluation Rubric

Evaluation Domain	Weight	Key Focus Criteria
Agent Design & Function Calling	**30%**	Accuracy in tool routing, state persistence across turns, edge-case recovery.
Zoho CRM Integration	**25%**	Robust OAuth 2.0 token management, accurate data mapping across Leads, Deals, and Cases.
Automotive Domain Prompting	**20%**	Professional OEM brand tone, accurate vehicle jargon, structured gathering of service parameters.
Latency & UI/UX Experience	**15%**	Low conversational latency, graceful error handling, streaming responses (or voice barge-in handling).
Code Quality & Documentation	**10%**	Modular code architecture, clean environment variable isolation, unambiguous setup instructions.